

net changes in total headcount.⁹

These studies highlight that pre-2023 adoption of AI technology often entailed internal reallocation rather than aggregate job loss. In contrast, our paper provides U.S. evidence on the implications of firm-level adoption during the first years of widespread *Generative* AI diffusion (2023–2025). By focusing not only on overall labor demand but on within-firm seniority composition, we provide evidence that GenAI-adopting firms reduce junior employment while leaving senior employment unaffected.

The Effects of Generative AI on the Labor Market: Finally, a rapidly growing empirical literature examines the labor-market effects of GenAI. Experimental studies generally find that GenAI complements less-experienced workers by boosting their productivity. For example, [Noy and Zhang \(2023\)](#) show that access to ChatGPT substantially reduces completion time and improves output quality, with especially large benefits for lower-ability workers. [Brynjolfsson et al. \(2025b\)](#) similarly find that GenAI assistance in customer support raised productivity by roughly 14 percent on average, with the largest gains for novices. [Dell’Acqua et al. \(2023\)](#) report comparable improvements in consulting workflows. Moreover, [Cui et al. \(2025\)](#) find especially high productivity gains and adoption rates for less experienced software developers. Related field-experimental evidence in [Dell’Acqua et al. \(2025\)](#) shows that adding a GenAI copilot reshapes teamwork and the division of expertise, shifting routine cognitive work to the tool and reorienting human effort toward higher-level tasks. These findings are consistent with the view that GenAI can act as a “leveler,” narrowing productivity gaps between less and more experienced workers ([Autor, 2024](#)).

A second wave of studies uses large-scale labor-market data to track employment dynamics across occupations by GenAI exposure. Closest to our work, [Brynjolfsson et al. \(2025a\)](#) show that since the late-2022 debut of GenAI, employment of young entry-level workers (ages 22-25) in the most AI-exposed occupations fell by about 13 percent relative to trend, while more experienced workers in those occupations saw stable or ris-

⁹Two related studies examine the effects of *Generative* AI adoption. [Humlum and Vestergaard \(2025\)](#) provide evidence from Denmark, linking large-scale worker surveys on GenAI adoption to matched employer-employee data. Despite rapid adoption, they do not find effects of adoption on earnings and hours, suggesting that labor-market impacts in Denmark remain minimal. [Chen and Stratton \(2025\)](#) analyze firm-level adoption of GitHub Copilot and Cursor using detailed engineering workflow data, documenting effects on productivity, task allocation, and organizational collaboration.

3 Conceptual Framework

3.1 Overview

This section develops a simple formal framework to analyze the “seniority-biased” labor demand effects of GenAI. The objective is to explain why GenAI adoption may disproportionately reduce demand for junior workers and to clarify potential mechanisms underlying this pattern.¹¹ The framework is motivated by two broad patterns in the emerging evidence: micro-level experimental studies consistently find that GenAI disproportionately increases the productivity of less-experienced workers (Brynjolfsson et al., 2025b; Noy and Zhang, 2023; Dell’Acqua et al., 2023; Cui et al., 2025), whereas labor market evidence, including our results, suggests that GenAI adoption and exposure are associated with declines in junior employment (Brynjolfsson et al., 2025a; Azar et al., 2025; Klein Teeselink, 2025; Lodefalk et al., 2026).

The model examines how GenAI adoption reshapes the composition of labor demand between junior and senior workers in a partial-equilibrium setting. It highlights three mechanisms: a *displacement effect*, whereby automation removes tasks previously performed by juniors; a *productivity effect*, whereby higher productivity reduces labor required on surviving tasks; and a *reallocation effect*, whereby juniors expand into some previously senior tasks as their relative productivity rises. The main result is that when tasks are complementary or Cobb-Douglas ($\sigma \leq 1$), displacement and productivity effects reduce junior labor demand, while reallocation is the only offsetting force. Whether junior employment declines more than senior employment therefore depends on the relative strength of these forces.

3.2 Setup

We build on the task-based framework of Acemoglu and Autor (2011) and Acemoglu and Restrepo (2018). We consider a short-run partial-equilibrium setting in which firm output Y and wages are held fixed. The goal is to isolate how adoption of GenAI by a firm

¹¹This framework explains the *incidence* of GenAI across seniority levels—why juniors may bear the labor demand cost while seniors do not. It is distinct from, and complementary to, the forward-looking hiring model in Supplemental Appendix A.5, which explains the *timing* of the employment response—why firms adjust junior hiring early, before automation is fully realized.

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

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